

PARIKSHAT SINGH

Computer Science Student | Machine Learning & AI | C++ & Python
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PROFESSIONAL SUMMARY

2nd-year CS student with active AI/ML research at DRDO's SSPL Lab and international academic collaborations. Built production-grade RAG systems; contributing to a deep learning project with Kennesaw University, USA. Proficient in Python, NumPy, Pandas, scikit-learn; strong in linear algebra, probability, and statistics.

EXPERIENCE

Research Intern – AI/ML | DRDO, Solid State Physics Laboratory, New Delhi June 2026 – Present

- Developed an AI-powered news intelligence platform with real-time news aggregation, AI-generated summaries, article explanations, and personalized Telegram alerts.
- Engineered scalable backend services using Python, MySQL, Redis, asynchronous programming, and LLM APIs, enabling automated content processing and intelligent news delivery.

SKILLS

Languages: Python, C, C++, Java

ML & AI: Machine Learning, Deep Learning, RAG, scikit-learn, TensorFlow

Data & Math: NumPy, Pandas, Statistics, Probability, Linear Algebra, Matplotlib

Web & Tools: Flask, FastAPI, Next.js, ChromaDB, Git, GitHub

Core CS: Data Structures & Algorithms, OOP, Problem Solving

PROJECTS

ElectroMentor – Multi-Agent Corrective RAG AI Tutoring System Collab: Republic Polytechnic, Singapore

Python, FastAPI, Next.js, ChromaDB, Google Gemini API

- Designed a multi-agent Corrective RAG pipeline using NLP, vector embeddings (ChromaDB), and document-grounded retrieval to answer domain queries from academic PDFs; significantly reduced hallucination rate via citation-grounded responses.
- Implemented supervised evaluation loops, automatic MCQ generation, and adaptive tutoring; pipeline covers text extraction, chunking, embedding generation, and semantic search. Research paper submitted (co-author).

RGB Image Processing / Photo Editor – Full-Stack Web App

Python (Flask), React.js, NumPy, Pillow, Axios, Framer Motion, Three.js

- Built full-stack image editor: React.js frontend (drag-and-drop, component architecture, Framer Motion UI, Three.js particle background) with Flask REST backend; Base64 pipeline eliminates temp file storage.
- Implemented NumPy pixel-level algorithms: brightness/contrast (int16 overflow prevention), per-channel RGB modification, channel swap (false-color), and inversion; Lodash debounce + useRef lock prevents concurrent API conflicts.

Weather Intel – ML-Based Weather Analysis & Risk Prediction

Python (Flask), Pandas, NumPy, Matplotlib

- Applied IQR and Z-score anomaly detection on weather datasets; engineered multi-variable risk prediction model using temperature, humidity, rainfall, and wind feature with interactive dashboards.

ACHIEVEMENTS & RESEARCH

- Contributing to an AI & Deep Learning research project with Kennesaw University, USA — focused on advanced ML model development.
- Co-author of research paper on ElectroMentor (Multi-Agent Corrective RAG tutoring system) — submitted for review at college level.
- Internal Hackathon Winner – Smart India Hackathon 2025 (College Level); multiple competitive coding event wins.
- Certified in C and C++. Active on LeetCode — consistent DSA practice across arrays, trees, DP, and graphs.

EDUCATION

B.Tech Computer Science – Bharati Vidyapeeth's College of Engineering

2024 – 2028 (Expected)

Current CGPA: 8.7 / 10

Senior Secondary (Class XII – Science) – St. Colombo Public School

2024

Percentage: 84.4%